

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues [#105](#) / [#61](#)
2026-05-13

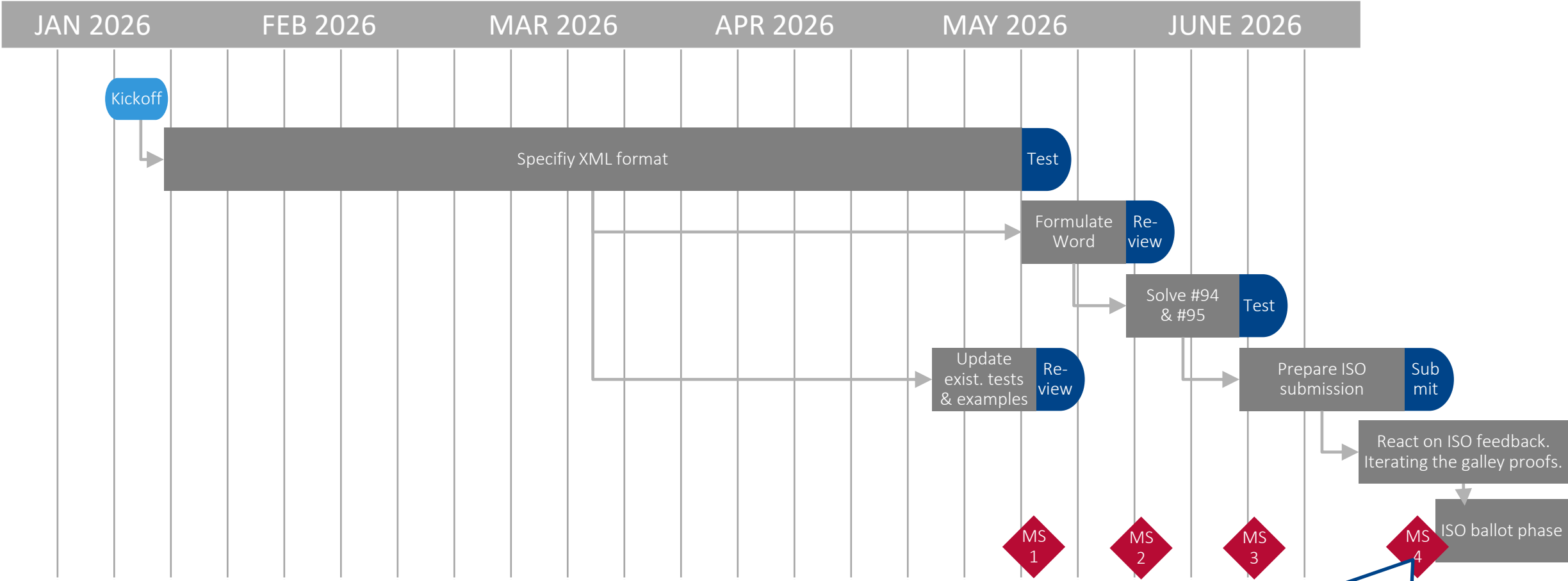
Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-05-13_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- Analysis: Which element attributes should (not) be available for the templates?
 - See SH's contribution of 2026-04-30.
- How to verify χ MCF 3.2 files via Schematron?
 - See NE's contributions of 2026-05-03.
- Steps we should take next



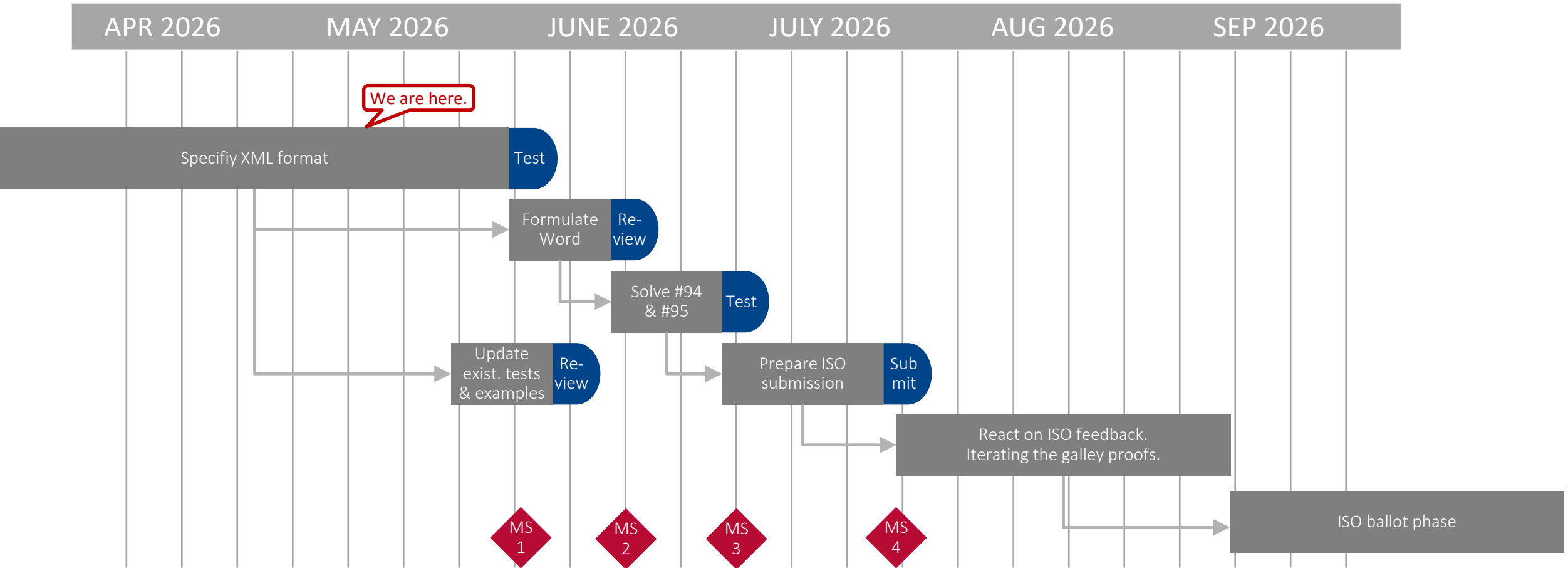
Timeline for the next release — as of 2026-04-29



Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.
Question was repeated at April 28th SC4/WG12 meeting. Though, no answer yet.



Timeline for the next release



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Conclusion of the Investigation

Combined XML verification by XML schema plus Schematron provides:

- Checks for several additional aspects, e.g. type confusion.
- Chance to separate business logic ($\mapsto$ Schematron) and formal syntax ($\mapsto$ Schema)

⇒ There are indisputable technical reasons to switch χ MCF 3.2 validation to a Schematron-based method.

However:

- Implementing the complete Schematron file will take weeks.
- Many checks in XML schema need to be reduced from “mandatory” to “optional”. Will also take time.

NE & SH volunteer for both these items.

CF will care for updating the existing χ MCF 3.2 example files. — Pending.

Analysis: Which element attributes should (not) be allowed for the templates? (SH, continued from 2026-04-29)

- See file "[Attributes invalid for templates.pdf](#)", as committed to GitHub 2026-04-30 09:13 UTC+2:00.



<threaded_connection/>, <nut/> & <washer/> templates

Which sub-elements are allowed in a template? (NE)

Discussion of status of <threaded_connection/> templates.

- Make <nut/> and <washer/> appear in the <connection_templates/> section.
Confirmed.
- Do we want <threaded_connection/> to appear in the <connection_templates/> section as well?
Confirmed: yes.
- However, <connection_0d/> shall be *not* allowed in <connection_templates/> section.
- Should <rivet/> subtypes (<solid/>, <self-piercing/>, etc.) appear in the <connection_templates/> section?
Decision: Only <rivet/> is allowed, not its sub-elements.
- Heat stakes, clips etc. are to be investigated later.



Directions (normal & tangential) in templates?

Decision:

- No directions shall appear inside a connection template.

Rationale:

- The combination of several connection templates with a number N of different directions delivers a high number (Cartesian product) of combined templates to be managed and hence cuts down the desired repetition count by the corresponding factor N (on average).
Consequently, this approach is neither scalable nor cost-effective.
- Idea (investigation postponed): Introduce *separate* templates for certain directions, e.g. “**negative_z**”, *independent* from connection templates.

- Also to be investigated later: Introduction of templates for `<custom_attributes/>`.

Recent results wrt. Schematron verification and `<threaded_connection>`.

See NE's commits of May 3, 2026:

- [split the gumdrop_template.xml into 2 files](#)
- [schematron checks against `<threaded\_connection/>`](#)
- [templates for rivet & bolt are introduced](#)

The following elements are introduced in `<connection_templates/>`:

`<rivet/>`

`<washer/>`

`<nut/>`

`<threaded_connection/>`

Any attributes of these elements that were 'required', will now be checked by the schematron file, not by the xsd.

- [Update README.md](#)

Next meetings

- 2026-05-20, 15:00 CEST — focus group on [#105](#) / [#61](#) — objective: agree on XML suggestions & examples → #106

Wed. 2026-05-27	Wed. 2026-06-03	Wed. 2026-06-10	CF on business trips. — Substitute dates:
—	Mon. 2026-06-01, 16:00	Fri. 2026-06-12, 16:00	... always CEST

- 2026-06-01, 16:00 CEST — focus group on [#105](#) / [#61](#) — objective: agree on XML suggestions & examples
- 2026-06-12, 16:00 CEST — focus group on [#105](#) / [#61](#) — objective: t.b.d.

CF sent / will send out invitations.

